

Rebuilding the Social Engine

Data recovery, cleaning and exploratory analysis of the corrupted Social Engine intake dataset.

12,000

unique, clean posts (from 12,360 raw rows)

9

corruption types detected and fixed

15,730

value-level changes, each logged with a rule

0

values imputed or invented

53 / 53

automated validation checks passed

8

statistically tested insights

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Repository: <https://github.com/panshularora/eda>

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Generated by `src/build_report.py` from pipeline outputs; every figure and number is reproducible with `python run_all.py`.

1 Executive summary

- **Recovered the exact dataset.** Following the recovery-log clue to `node_07` through the recovery shell yielded `Social_Engine_Posts_Corrupted.csv` (12,360 × 8) and `Social_Engine_Users.csv` (1,500 × 5). SHA-256 hashes are pinned in the pipeline, so any altered input is detected.
- **Found nine corruption types, including four that are easy to miss:** timestamps that lost their time of day (3,526 posts), two different spellings of missing values, five kinds of injected text suffixes (`&` , `<br>` , `<div>` , `Ã©` , trailing newlines), and `NULL` tokens hidden behind those suffixes.
- **Cleaned without inventing data.** Values are corrected only where the original is deterministically recoverable. Unrecoverable values stay `NULL`, since they are missing completely at random (every χ^2 test $p \geq 0.145$). Imputation would add false spikes and bias Phase 2 correlations.
- **Avoided the “midnight spike” trap.** Padding date-only timestamps with 00:00:00 would put 32.5% of all posts at midnight ($11.1 \times$ any other hour). Real posting activity is flat across the day ($p = 0.475$).
- **What the data says:** volume is stationary (~33 posts/day), engagement metrics are independent and uniform, no platform has a real engagement edge, followers do not predict engagement (Spearman $p = 0.011$), totals simply track post count (Spearman $p = 0.841$), user language is unrelated to location, and seasonal campaigns are not mentioned in season. Each claim is backed by a test in Section 7.

2 Recovering the dataset

The rulebook withholds the dataset; it had to be recovered from the corrupted Social Engine site. Our path:

- 1 The landing terminal boots and reports `data_intake.service ... FAILED`. *Enter Recovery Mode* asks for a locally stored operator identity.
- 2 All ten dashboard modules fail in different ways (hang, signal lost, page error, access denied, no response). The ARCHIVE module shows `ARCH_LINKLOST_07`. After a few failures the site hints: *“Nothing responding? There may be another way in.”*
- 3 Each **system log** line **begins** with a timestamp and the component reporting it. Only one component is healthy: `node_07 responded 200 (intermittent)`, then `watchdog last known good node: node_07` and `dashboard link to node_07: severed`.
- 4 The recovery shell (terminal button) accepts `help`, `status`, `scan`, `logs`. `scan` confirms *“node_07 responding intermittently, outside normal routing. Manual reconnection may be possible.”*
- 5 `connect node_07` stabilises the link and opens **ARCHIVE NODE 07**. We recovered both files listed under *Dataset 01*.

Integrity note. The site builds both CSVs in the browser and downloads them as Blobs, so they never show up as separate requests in the Network tab. We confirmed our downloads match the copies embedded in the site bundle byte-for-byte. SHA-256: posts `49a460d6b6d8b7f5...`, users `d30efe470a4dd561...`. A `.gitattributes` rule keeps git from rewriting their line endings, so the hashes still match after a clone on any OS.

3 Dataset overview

Table	Grain	Raw rows	Clean rows	Columns (raw)
Posts	one post	12,360	12,000	<code>post_id</code> , <code>user_id</code> , <code>platform</code> , <code>text_content</code> , <code>timestamp</code> , <code>likes</code> , <code>shares</code> , <code>comments</code>

Table	Grain	Raw rows	Clean rows	Columns (raw)
Users	one account	1,500	1,500	<code>user_id</code> , <code>location</code> , <code>language</code> , <code>account_created</code> , <code>follower_count</code>

The posts cover **2024-05-01 → 2025-04-30** (365 days) across 5 platforms, written by 1,500 users in 33 cities and 19 countries using 10 languages. All users signed up during 2023 (2023-01-01 → 2023-12-31). No post predates its author's account. Every post's `user_id` exists in Users and every user has at least one post, so referential integrity holds in both directions. **Only 7,366 posts (61.4%) have platform, text and likes all present.** Dropping incomplete rows would therefore discard 39% of the data. We keep every row and let each analysis use the fields it needs.

4 Corruption audit & fixes

Every cell is read as raw text (`keep_default_na=False`), so nothing is silently converted. Each corruption is detected by an explicit rule, its **evidence is computed and asserted in code** (the pipeline stops if an assumption breaks), then the fix is applied and logged. Counts are shown for the raw file and after de-duplication, so they reconcile exactly.

#	Corruption	Evidence (computed)	Action	Raw file	After dedup
1	Exact duplicate rows	352 <code>post_ids</code> appear 2–3 times (8 triplicates); every copy identical in all 8 columns	Keep first occurrence, drop the rest	360	—
2	Timestamp as Unix epoch	10-digit integers; decoded as UTC they fall in the same 2024-05-01 → 2025-04-30 window as ISO values	Convert to UTC <code>YYYY-MM-DD HH:MM:SS</code>	3,788	3,669
3	Timestamp as dd-mm-yyyy (time of day lost)	2nd field never > 12, 1st field > 12 in 2,113 rows ⇒ day-first; no time component exists	Keep <code>post_date</code> ; <code>post_datetime_utc</code> = NULL; <code>timestamp_precision</code> = 'day'	3,622	3,526
4	Missing platform, two spellings	" and 'NULL' in a ~2:1 ratio (1,177 : 607)	Unify to NULL. Not imputed (no signal predicts it)	1,846	1,784
5	Missing likes, two spellings	" and 'NULL' (1,199 : 615)	Unify to NULL. Not imputed	1,858	1,814
6	Negative likes (sign flip)	All 509 negatives are float-formatted (<code>-2388.0</code>) while every other value is an integer; <code> negative </code> has the same distribution as positives (KS $p = 0.626$)	<code>abs()</code> , cast to integer, flag <code>dq_likes_sign_corrected</code>	525	509
7	Missing text, two spellings	" and 'NULL' (1,154 : 534), plus 91 'NULL' tokens hidden behind an artifact (e.g. <code>NULLÃ©</code>)	Unify to NULL (no placeholder text)	1,840	1,779
8	Text suffix artifacts (5 kinds)	<code>&amp;</code> 328, <code><div></code> 335, <code>
</code> 311, <code>\n\n</code> 329, mojibake <code>Ã©</code> 306. Each occurs only as the final characters, never mid-text, at most one per post	Strip the suffix; flag the kind in <code>dq_text_artifact_removed</code>	1,657	1,609

#	Corruption	Evidence (computed)	Action	Raw file	After dedup
9	Double spaces inside text	Gaps where a template slot was empty (e.g. "recommend. #Beauty")	Collapse runs of whitespace to one space	—	680

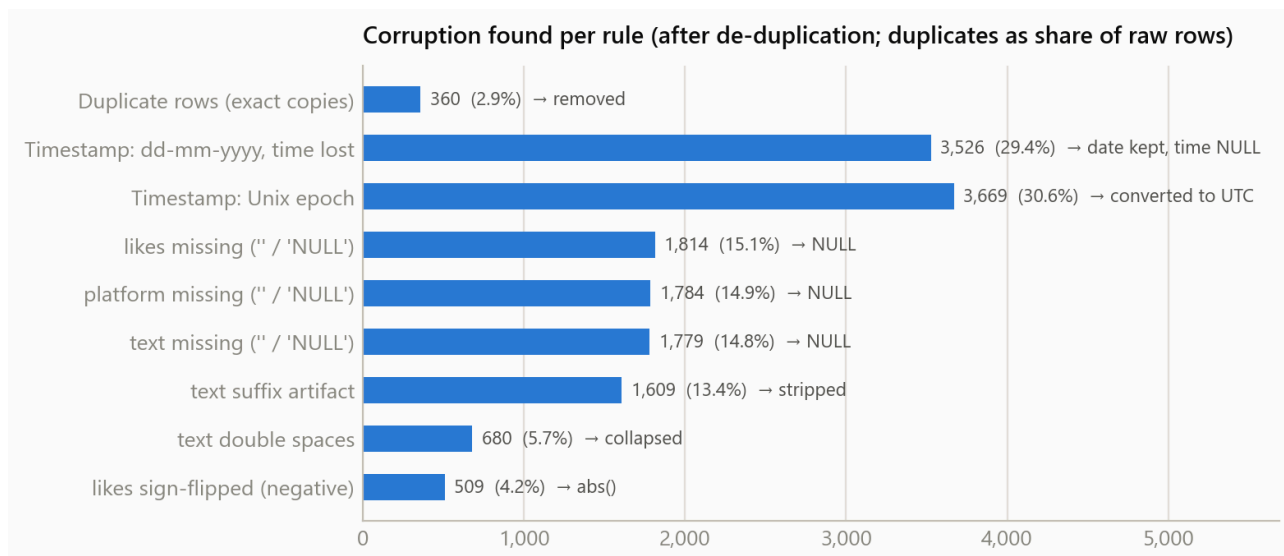


Figure 1. Size of each corruption after de-duplication. Percentages are shares of the 12,000 unique posts; duplicates are a share of raw rows.

Lineage. `reports/change_log.csv` records all 15,730 changed values as (post_id, column, rule, old value, new value). The cleaned posts table also carries three data-quality flags (`dq_duplicate_copies_removed`, `dq_likes_sign_corrected`, `dq_text_artifact_removed`), so corrections stay queryable in SQL. **Users** passed every check (no missing, padded, duplicate or malformed values), so it needed no corrections. We only added `city`, `country` and `language_name` for grouping. ISO timestamps were already correct and were only re-rendered in the common output format.

5 Decisions, assumptions & alternatives

5.1 Date-only timestamps keep their date and get no invented time

3,526 posts (29.4%) carry only a date. The obvious conversion (`pd.to_datetime` → `2024-09-25 00:00:00`) invents a time. The cleaned table therefore has `post_date` (always present), `post_datetime_utc` (NULL when unknown), `timestamp_precision` and `timestamp_source_format`.

The “midnight spike” is created by the cleaning step, not by users

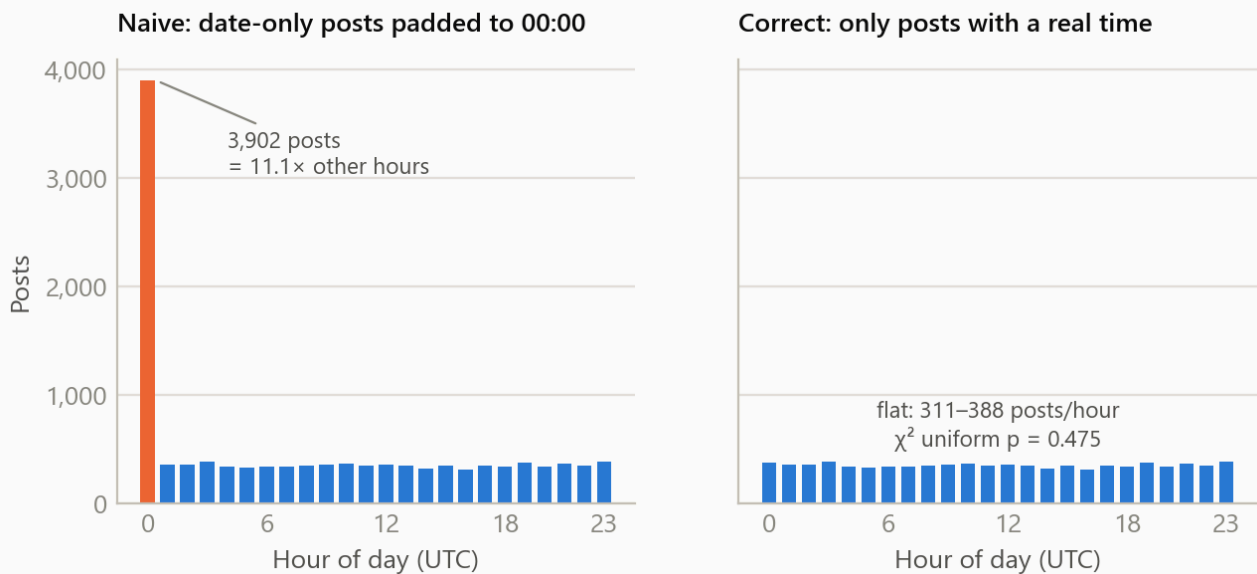


Figure 2. Left: the hour-of-day profile if date-only posts are padded to midnight. Right: the true profile using only posts with a real time.

5.2 Negative likes are sign flips; missing likes stay NULL

Two independent pieces of evidence show a sign-flip corruption rather than “dislikes”. First, the format: every negative value is a float string while every other value is an integer. Second, the distribution: the absolute values of negatives match positives (KS $p = 0.626$; means 2,460.7 vs 2,493.5). Missing likes cannot be recovered. They correlate with nothing, so any model reduces to the median, and median imputation would put 1,817 posts on one value (2,498) and shrink the standard deviation by 7.9%.

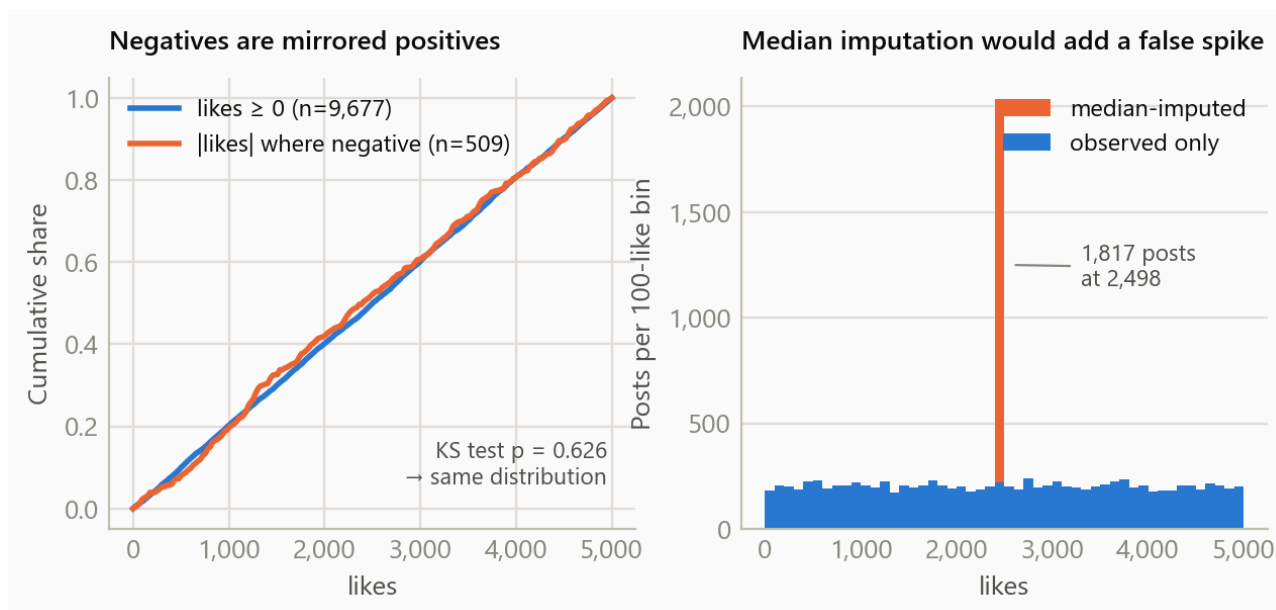


Figure 3. Left: the cumulative distribution of |negative likes| overlaps positive likes. Right: median imputation would create a spike that later analysis could mistake for an anomaly.

5.3 Missing values are missing completely at random

We tested each corruption flag against platform, timestamp source format and month (17 χ^2 tests). None is significant; the smallest p -value is 0.145. Missingness in one column does not predict missingness in another: likes and platform are both missing in 260 posts vs 270 expected under independence, likes and text in 282 vs 269. Because the data is MCAR, analyses that simply skip missing values are unbiased, and filling values in would add nothing but false precision.

Corruption rate by platform is flat: every χ^2 test is non-significant (MCAR)					
likes missing (p = 0.604)	15.4%	14.9%	14.2%	15.8%	15.7%
text missing (p = 0.957)	15.0%	15.1%	15.5%	15.0%	14.6%
likes sign-flipped (p = 0.154)	3.4%	4.8%	3.9%	4.5%	4.0%
text artifact (p = 0.413)	14.0%	14.3%	13.1%	12.4%	13.6%
duplicated on ingest (p = 0.495)	2.9%	2.5%	2.7%	3.4%	2.9%
	Facebook	Instagram	Reddit	Twitter	YouTube

Figure 4. Corruption rate per platform (rows with a known platform), with χ^2 p-values. Flat rows mean the corruption is independent of platform.

5.4 Assumptions register

#	Assumption	Support	If wrong
A1	Unix epochs are UTC; naive ISO strings use the same UTC clock	Epoch is UTC by definition. Both formats span the same window (2024-05-01 → 2025-04-30) with matching flat hour profiles	Hour-of-day shifts by a constant offset; daily and monthly results barely change
A2	<code>dd-mm-yyyy</code> is day-first	Proven: month field max = 12, day field max = 31	The pipeline raises an error
A3	Negative likes are sign-flipped, not downvotes	Format fingerprint plus identical distribution (5.2)	509 posts (4.2%) would carry the wrong sign; flagged, so they can be excluded
A4	The five suffixes are injected, not user-written	Asserted: never mid-text, never stacked, only at the end. Decoding <code>&amp;</code> or <code>Ã©</code> would leave a stray <code>&</code> or <code>é</code> glued to punctuation	The pipeline raises an error if a suffix appears mid-text
A5	Duplicate post_ids are repeated ingestions of one post	Asserted: every copy identical in all columns	The pipeline raises an error on conflicting copies
A6	Template oddities (“How do I fix about...”, “Pepsi Pepsi Zero Sugar”) are original content	They occur at template scale, not at corruption rates, and are not suffix-like	Rewriting them would be fabrication, so they are kept verbatim

5.5 Alternatives we rejected

Alternative	Why rejected
Mode-impute platform	With five near-equal platforms the mode is right only ~20% of the time. It would push Facebook from 17.3% to 32.1% of posts (3,858) and fabricate a “dominant platform”.
Median-impute likes	Creates a 1,817-post spike, shrinks variance and biases correlations toward 0 (Figure 3).
<code>html.unescape()</code> the text	Fixes only 1 of 5 artifact kinds and turns <code>&amp;</code> into a stray trailing <code>&</code> . <code>
</code> , <code><div></code> , <code>Ã©</code> and newlines would survive.
Placeholder text such as “[CONTENT UNAVAILABLE]”	Pollutes word counts and text search, and SQL cannot tell it apart from real content. NULL is the honest encoding.
Drop incomplete rows	Discards 39% of posts for no bias reduction, since the data is MCAR.

Alternative	Why rejected
"Fix" language to match location	The true language is unknowable. The mismatch is itself a finding (Insight 7).

6 Validation

`src/validate.py` re-reads the written CSVs and asserts **53 contracts; 53 pass**. They cover schema and key uniqueness, the foreign key, no leftover missing-value tokens, platform vocabulary, date window and precision consistency, integer ranges (likes 0–5,000, shares 0–2,000, comments 0–1,000), zero HTML, mojibake, newlines or double spaces left in text, and no post before account creation. It also **re-runs the full pipeline and confirms the outputs are byte-identical**. The full table is in `reports/validation_report.md`.

7 Exploratory analysis

Conventions: we report the test and p-value for every pattern. $p \geq 0.05$ is reported as "no evidence", never as a finding. Likes-based statistics use the 10,186 posts with known likes. Hour-of-day uses the 8,474 posts with a real time.

7.1 Volume over time

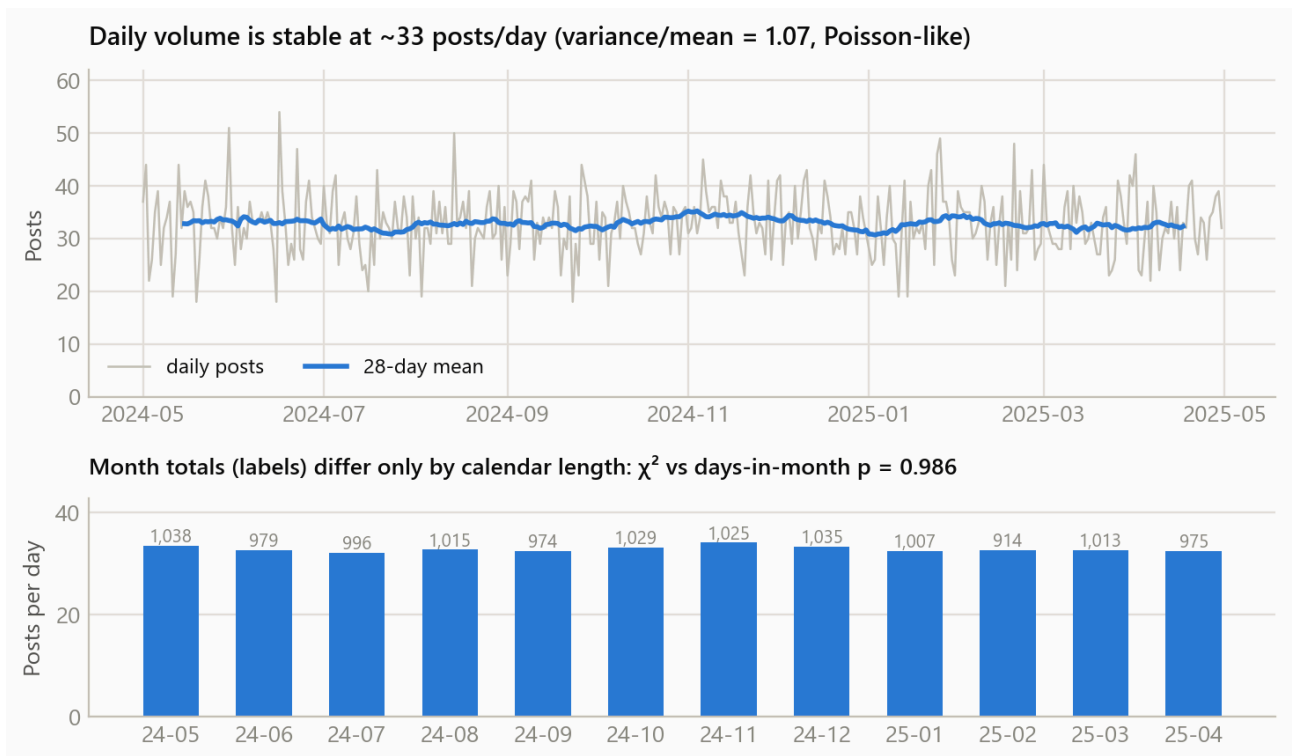


Figure 5. Daily posts with a 28-day mean (top) and posts per day by month, labelled with monthly totals (bottom).

Volume is flat at **32.9 posts/day** (range 18–54). Variance/mean is 1.07, i.e. Poisson noise around a constant rate. The largest month (2024-05, 1,038) and smallest (2025-02, 914) differ only by calendar length. Per-day rates run 32.1–34.2, and a χ^2 test of monthly totals against days-in-month gives $p = 0.986$. Day of week is also flat ($p = 0.722$). **No single day is anomalous** after correcting for 365 tests (Poisson tail, $\alpha = 0.05/365$). Only 2 days exceed $z = 3$, against 0.74 expected by chance.

7.2 Engagement distributions

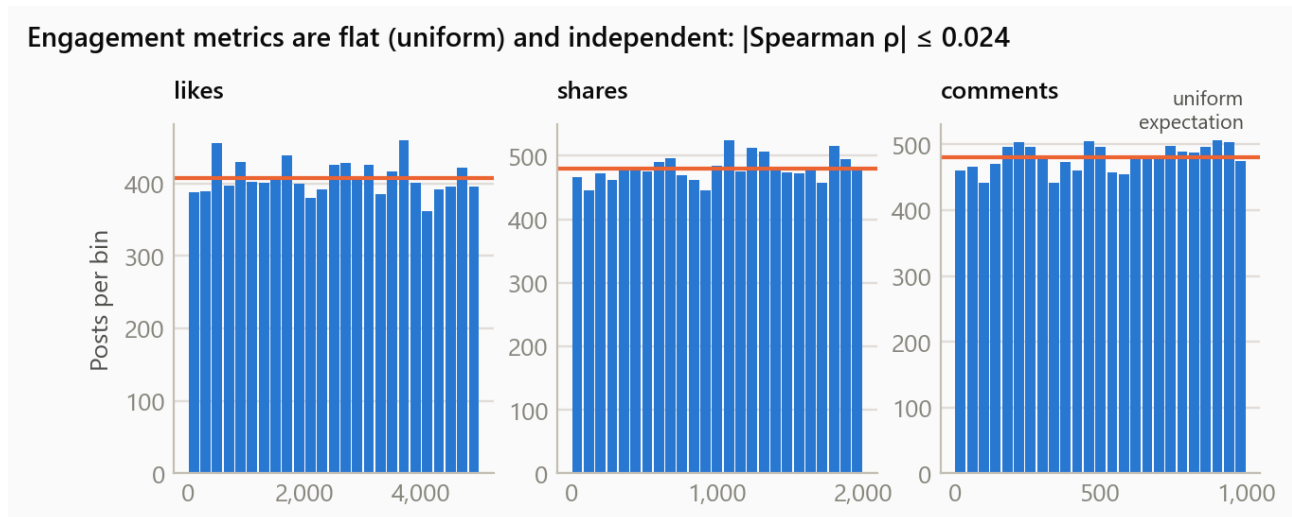


Figure 6. Histograms of likes, shares and comments; the orange line is the count expected under a uniform distribution.

Likes, shares and comments are each **uniform** on $[0, 5,000]$, $[0, 2,000]$ and $[0, 1,000]$ (KS vs uniform: $p = 0.407$, $p = 0.173$, $p = 0.195$), and **mutually independent** (Spearman ρ : likes–shares -0.001 , likes–comments 0.01 , shares–comments 0.024). The means (2,491.9 / 1,007.2 / 504.3) sit at the midpoints, so expected engagement splits about 5 : 2 : 1. There is no heavy tail and no viral outlier: the maxima are hard caps.

7.3 Platforms

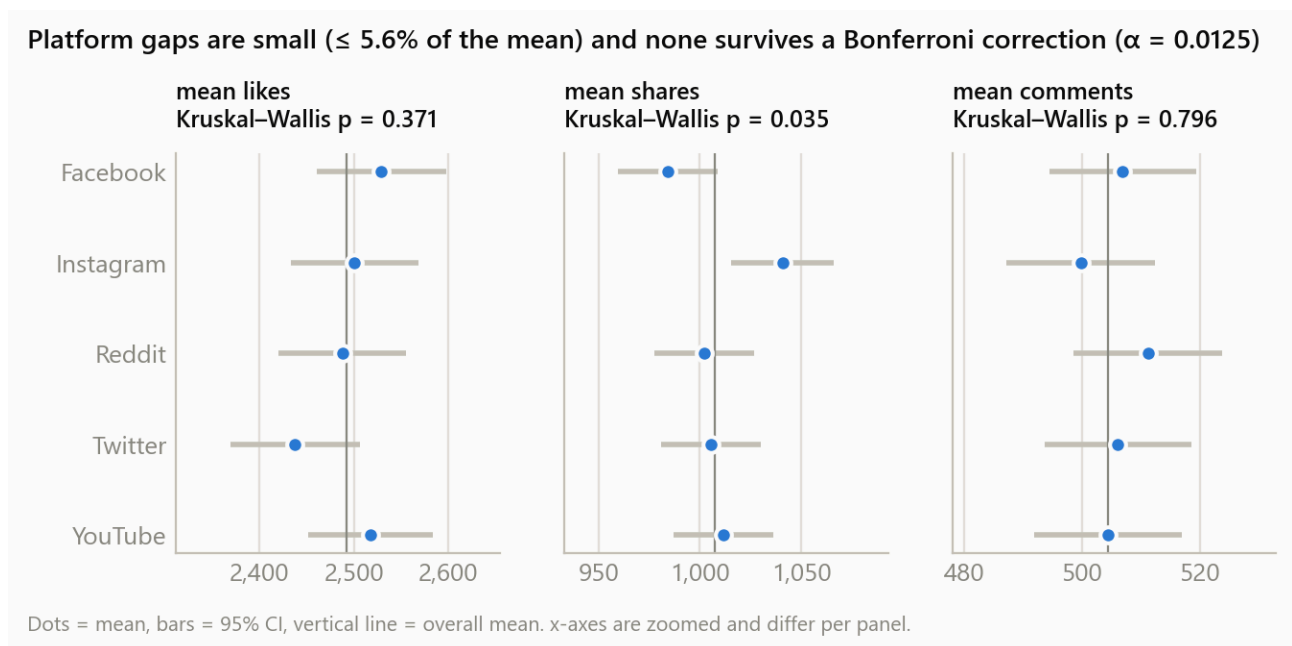


Figure 7. Mean engagement per platform with 95% confidence intervals.

Platform volumes are balanced (Facebook 2,074, YouTube 2,073, Twitter 2,049, Reddit 2,031, Instagram 1,989; largest vs smallest 4.3%, χ^2 vs uniform $p = 0.658$). Mean likes differ by at most 3.7% of the overall mean ($p = 0.371$) and comments by 2.3% ($p = 0.796$). Shares are the only nominal difference: Instagram 1,040.8 vs Facebook 984.2, $p = 0.035$. That does not survive a Bonferroni correction for four metrics ($\alpha = 0.0125$), and the effect size is negligible ($\epsilon^2 = 0.001$).

7.4 Users: activity, reach and engagement

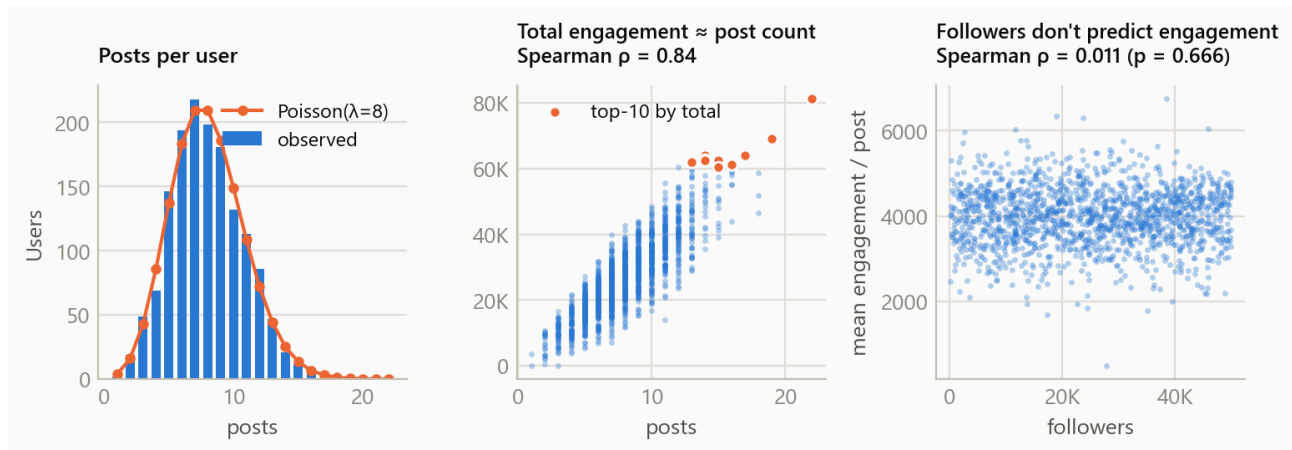


Figure 8. Posts per user against a Poisson(8) curve; total engagement vs post count; mean engagement per post vs followers.

Posts per user follow a **Poisson($\lambda = 8.0$)** distribution (variance/mean 1.02, range 1–22). 1,479 users (98.6%) post on 2+ platforms, averaging 3.73. A null model that assigns each post a random platform predicts 1,475 and 3.71, so cross-platform posting is not a behaviour, just chance. Followers (uniform 109–49,944, KS $p = 0.870$) do not predict engagement per post ($p = 0.011$, $p = 0.666$) or activity ($p = -0.01$).

Ranking users by **total** engagement mostly ranks post counts ($\rho = 0.841$). The top 10 by total average 15.9 posts, twice the norm, and **none of them** appears in the top 10 by **mean engagement per post** (minimum 5 posts with known likes). The per-post leaders are:

#	user_id	posts	mean engagement / post	followers	location
1	user_py6fq7ii	6	6,339.6	18,940	Delhi, India
2	user_mcchsnr8	7	5,969.5	2,602	Munich, Germany
3	user_3318o2cp	7	5,713.3	2,374	Cairo, Egypt
4	user_5s9ifp0y	11	5,566.3	28,412	Barcelona, Spain
5	user_0e2how9t	6	5,538	23,563	Melbourne, Australia

7.5 Language and location

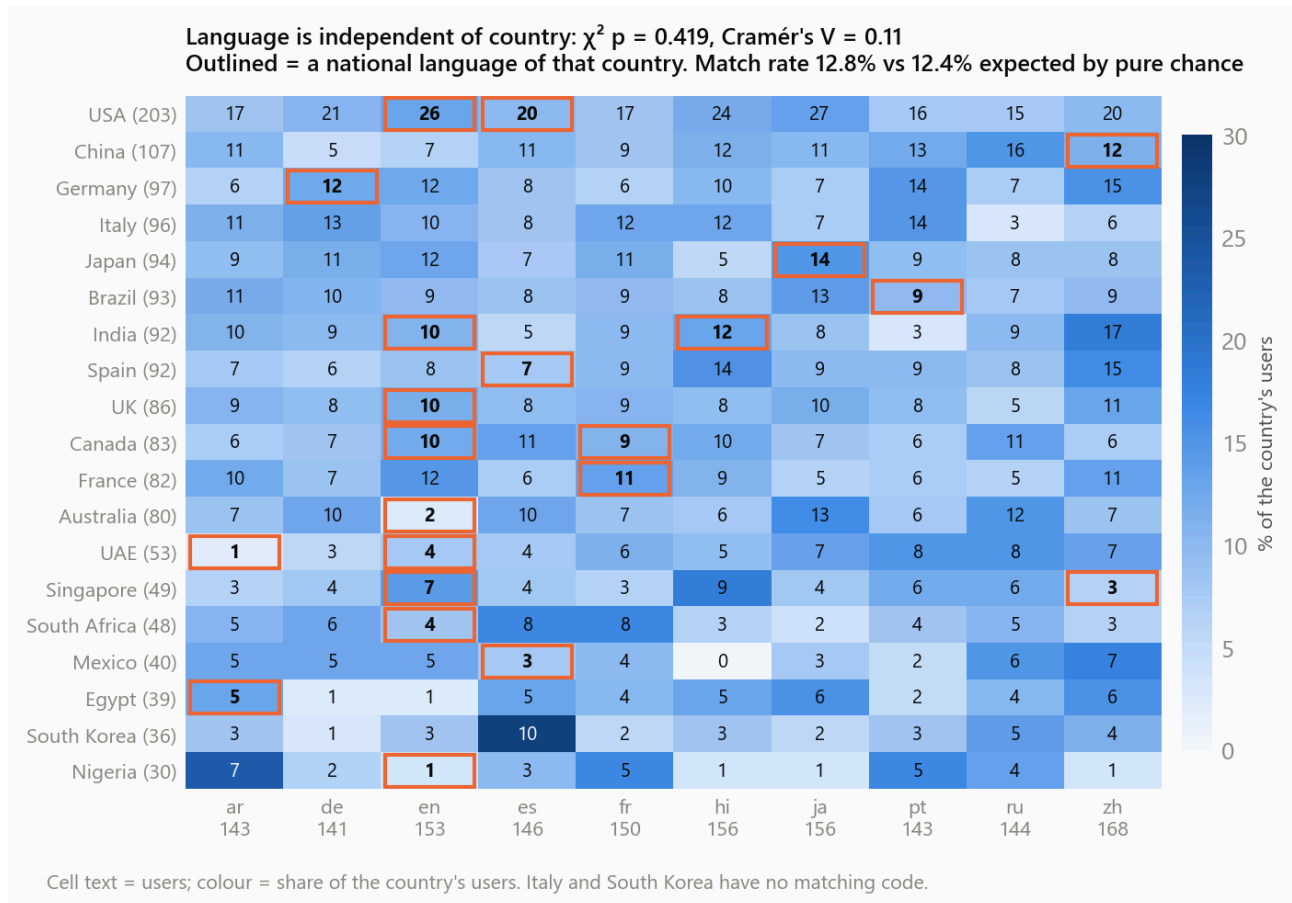


Figure 9. Users by country × language. Colour is the share of each country's users; outlined cells are a national language of that country.

The ten languages are equally common ($p = 0.895$ vs uniform) and **statistically independent of country** ($\chi^2 p = 0.419$, Cramér's $V = 0.111$). Only 12.8% of users speak a national language of their country, against 12.4% expected from random assignment (binomial $p = 0.667$). Examples: Berlin, Germany → ja, Munich, Germany → zh, São Paulo, Brazil → ar. The 132 users in Italy and South Korea cannot match at all, since it and ko are absent. The most represented country is the USA (203 users).

7.6 Text content

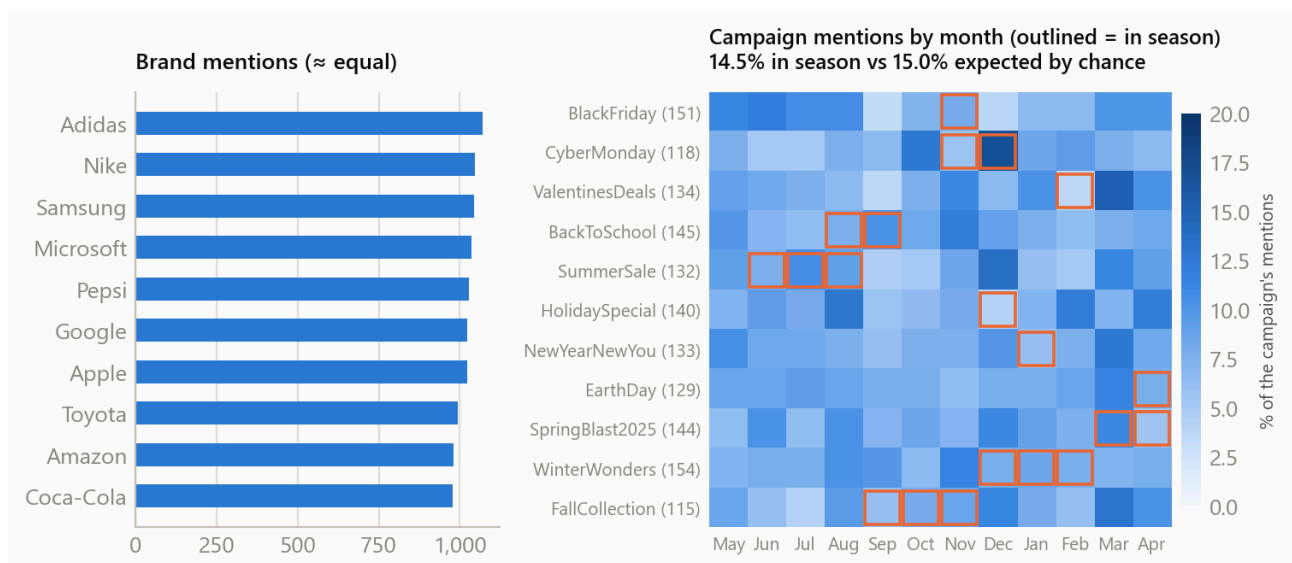


Figure 10. Left: posts mentioning each brand. Right: when each seasonal campaign is mentioned; outlined cells are its real season.

The 10,221 texts are near-unique (10,219 distinct) but built from templates: six openers such as “Just unboxed my new...”, “Just saw an ad for...” and “Comparing...” each start 980–1,052 posts. Mentions spread evenly over 10 brands (977–1,070 posts each), 29 hashtags (2.01 per post) and 15 @handles (in 13.5% of posts). Verdict phrases split into positive 3,268, negative 3,306 and neutral 1,581. Engagement does not differ by verdict ($p = 0.221$). **40.9% of posts that state an emotion contradict their own verdict**, e.g. “*Bummed out with my new Air Max from Nike! Absolutely loving it. #Travel, #CustomerService, #Fashion*”. Seasonal campaigns land in their season 14.5% of the time vs 15.0% by chance; *SpringBlast2025* even appears in 96 posts dated 2024.

8 Key insights

1 • The corruption is machine-injected at fixed rates, uniformly at random data quality

~15% missing in each of platform, likes and text; 13.41% text suffixes (five kinds of ~2.7% each); 4.24% sign flips; 2.93% duplicates. All 17 dependence tests are non-significant, co-missingness matches independence, and each missing column mixes `' '` and `NULL` at the same ~2:1 ratio. That points to a single random corruption process, not a failing source.

So what / Phase 2: Available-case SQL (`WHERE Likes IS NOT NULL`) is unbiased. Keep NULLs; do not `COALESCE` to a median.

2 • The “midnight posting spike” is an artifact of bad parsing anomaly

29.4% of timestamps have no time of day. Defaulting them to 00:00 makes hour 0 hold 3,902 posts (11.1× any other hour). With real times only, posting is flat across all 24 hours (311–388 per hour, $p = 0.475$).

So what / Phase 2: Any hour-of-day query must filter `timestamp_precision = 'second'`; day, week and month queries can use `post_date` for all rows.

3 • Activity is stationary: no trend, seasonality or anomalous day trend

32.9 posts/day with Poisson noise. Monthly totals are explained by days-in-month ($p = 0.986$), weekdays are flat ($p = 0.722$), and no day survives a Bonferroni-corrected Poisson test. The “May peak” and “February dip” are 31-day and 28-day months.

So what / Phase 2: Normalise counts by days (posts/day) before comparing periods, and test spikes against Poisson noise before calling them anomalies.

4 • Likes, shares and comments are independent uniform variables correlation

Uniform on fixed caps ($KS\ p \geq 0.17$), $|p| \leq 0.024$, no long tail. Real social data is heavy-tailed and positively correlated, so this looks like synthetic generation. A post with many likes is no more likely to be shared.

So what / Phase 2: Expect near-zero correlations. Define “high engagement” by percentile (e.g. top 10%) rather than absolute thresholds, and use rank-based statistics.

5 • No platform has a real engagement advantage grouping

Gaps are $\leq 5.6\%$ of the mean. The one nominal result (shares, Instagram vs Facebook, $p = 0.035$) fails multiple-testing correction, with $\epsilon^2 = 0.001$.

So what / Phase 2: Platform-level rankings will be driven by noise. Report confidence intervals alongside any `GROUP BY platform` result.

6 · Volume, not reach, drives total engagement behaviour

Followers show no relationship to per-post engagement ($p = 0.011$). Total engagement tracks post count ($p = 0.841$), and the top-10 by total and top-10 by per-post average share **zero** users. Cross-platform posting (98.6% of users) matches a random-assignment null model.

So what / Phase 2: For behavioural grouping, rank by per-post rates with a minimum-activity threshold (window functions such as `NTILE / PERCENT_RANK`), not by `SUM()`. “Influencer” cannot be inferred from `follower_count`.

7 · User language is not a location signal consistency

Language is independent of country ($V = 0.111$), with a national-language match of 12.8% vs 12.4% by chance. 132 users live in countries whose language is not coded at all.

So what / Phase 2: Treat language and country as separate dimensions. Do not derive regional language claims (e.g. “Asia-Pacific language dominance”) from this field.

8 · Post text is templated and carries no temporal or engagement signal text

Brands, hashtags and verdicts are evenly balanced. 40.9% of emotional posts contradict their own verdict, engagement does not vary by sentiment ($p = 0.221$), and campaigns are mentioned out of season (e.g. `SpringBlast2025` in 2024).

So what / Phase 2: Text features (brand, campaign, sentiment) suit descriptive counts only. Do not expect them to explain engagement or timing.

9 Limitations & Phase 2 readiness

- **Unrecoverable values remain NULL:** platform 14.9%, likes 15.1%, text 14.8%, time of day 29.4%. This is a deliberate trade of completeness for correctness.
- **Timezone:** A1 cannot be proven for naive ISO strings. Hour-of-day findings would shift by a constant if the source clock were not UTC.
- **Sentiment** relies on the dataset's fixed verdict phrases, not a general NLP model. It is exact for these templates but not transferable.
- **SQL-ready:** the cleaned CSVs load directly into a two-table schema (`users` 1—N `posts`) with typed columns, primary and foreign keys, and CHECK constraints that mirror our validation (`sql/schema.sql`, loaded by `src/build_sqlite.py`).

10 Reproducibility

Step	Command	Output
Environment	<code>pip install -r requirements.txt</code>	pandas 2.2.2, numpy 1.26.4, scipy 1.17.1, matplotlib 3.10.9
Everything	<code>python run_all.py</code>	Runs the six steps below in order (~15 s on a laptop)
1 Clean	<code>python src/clean.py</code>	<code>data/cleaned/*.csv</code> , <code>reports/cleaning_audit.json</code> , <code>reports/change_log.csv</code>
2 Validate	<code>python src/validate.py</code>	<code>reports/validation_report.md</code> (exits non-zero on failure)

Step	Command	Output
3 JSON	<code>python src/export_json.py</code>	<code>data/cleaned/social_engine_cleaned.json</code> (both tables in one file)
4 EDA	<code>python src/eda.py</code>	<code>reports/figures/*.png</code> , <code>reports/eda_stats.json</code>
5 SQL	<code>python src/build_sqlite.py</code>	<code>data/social_engine.db</code> + SQL cross-checks
6 Report	<code>python src/build_report.py</code>	This document (HTML + PDF)

Walkthrough notebook: `notebooks/Phase1_Walkthrough.ipynb` (executed, with outputs). Column definitions: `data/cleaned/DATA_DICTIONARY.md`. Decision log: `docs/CLEANING_DECISIONS.md`. Output hashes: posts `01c8373c048b97fb...`, users `9e7c1e650dd86984...`.